2-adic Valuations of $3n + 1$ by Residue Class mod 8

Speculative Results Series, Paper 66

Jon Seymour
jon@wildducktheories.com
Sydney, Australia

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Abstract

We prove that the 2-adic valuation $v_2(3n+1)$ under the Syracuse map $S(n) = (3n+1)/2^{v_2(3n+1)}$ is exactly determined — or has an exactly determined expectation — by the residue class of n modulo 8. Specifically, writing $c_r = E[v_2(3n+1) \mid n \equiv r \pmod 8]$:

r	c_r	Status
1	2	exact constant
3	1	exact constant
5	4	exact expectation
7	1	exact constant

For $r \in \{1, 3, 7\}$ the valuation is *constant* on the entire residue class — a direct consequence of the arithmetic of $3n+1$ modulo 8. For $r = 5$ the valuation varies, but its expectation is exactly 4: writing $n = 8k+5$ gives $v_2(3n+1) = 3 + v_2(3k+2)$, and $E[v_2(3k+2)] = 1$ follows from $\gcd(3, 2^j) = 1$, which makes $3k+2$ uniformly distributed modulo 2^j for every $j \geq 1$.

The sum of the four expected valuations is $c_1 + c_3 + c_5 + c_7 = 8 = 4 \times 2$, so the mean valuation across all four classes equals 2, the same as the constant valuation of class 1. Equivalently, $3^4/2^8 = 81/256 = (3/4)^4$: the product $\prod_r 3/2^{c_r}$ of the four asymptotic contraction ratios equals $(3/4)^4$, whose fourth root is exactly $3/4$.

This result provides the exact integer-exponent foundation for understanding the asymptotic contraction rates of the Syracuse map by residue class, and in particular why the 5 (mod 8) class — with $c_5 = 4$, giving asymptotic ratio $3/2^4 = 3/16$ — acts as a dynamical sink that drives orbits toward 1 with exceptional efficiency.

Status. This paper contains proved theorems. The main results (Theorem 2.1 and Corollary 3.1) are unconditional. The dynamical interpretation (Section 4) is stated as observation and conjecture.

Contents

1 Background and Notation

We work with odd positive integers throughout. The *2-adic valuation* $v_2(n)$ denotes the exponent of the largest power of 2 dividing n.

Definition 1.1 (Syracuse map)**.** The *Syracuse map* $S : \mathbb{O} \to \mathbb{O}$ on odd positive integers is

$$S(n) = \frac{3n+1}{2^{v_2(3n+1)}}.$$

The ratio $S(n)/n = (3n+1)/(n \cdot 2^{v_2(3n+1)})$ measures the multiplicative change per Syracuse step. Its asymptotic behaviour as $n \to \infty$ is governed entirely by the exponent $v_2(3n+1)$, since $(3n+1)/n \to 3$. The central question of this paper is: what is $v_2(3n+1)$, and how does it depend on n mod 8?

2 Main Theorem: Valuations by Residue Class

Theorem 2.1 (2-adic valuations of $3n+1$ mod 8)**.** *Let n be an odd positive integer and write $c_r = E[v_2(3n+1) \mid n \equiv r \pmod 8]$ where the expectation is over the uniform (natural) density on the residue class. Then:*

1. $n \equiv 1 \pmod 8$*:* $v_2(3n+1) = 2$ *for every such* n*, so* $c_1 = 2$.
2. $n \equiv 3 \pmod 8$*:* $v_2(3n+1) = 1$ *for every such* n*, so* $c_3 = 1$.
3. $n \equiv 5 \pmod 8$*:* $v_2(3n+1) \geq 3$ *for every such* n*, and* $E[v_2(3n+1) \mid n \equiv 5 \pmod 8] = 4$*, so* $c_5 = 4$.
4. $n \equiv 7 \pmod 8$*:* $v_2(3n+1) = 1$ *for every such* n*, so* $c_7 = 1$.

Proof. Write $n = 8k + r$ for $r \in \{1, 3, 5, 7\}$ and $k \geq 0$.

(1) $r = 1$**:** $3n + 1 = 3(8k + 1) + 1 = 24k + 4 = 4(6k + 1)$. Since $6k$ is even, $6k + 1$ is odd, so $v_2(6k + 1) = 0$ and $v_2(3n + 1) = v_2(4) + v_2(6k + 1) = 2 + 0 = 2$ for every k.

(2) $r = 3$**:** $3n + 1 = 3(8k + 3) + 1 = 24k + 10 = 2(12k + 5)$. Since $12k$ is even, $12k + 5$ is odd, so $v_2(3n + 1) = 1$ for every k.

(3) $r = 5$**:** $3n+1 = 3(8k+5)+1 = 24k+16 = 8(3k+2)$. Hence $v_2(3n+1) = 3+v_2(3k+2)$ for every k, establishing $v_2(3n + 1) \geq 3$. It remains to show $E[v_2(3k + 2)] = 1$.

For any $j \geq 1$, since $\gcd(3, 2^j) = 1$, multiplication by 3 is a bijection on $\mathbb{Z}/2^j\mathbb{Z}$. Therefore as k ranges uniformly over $\mathbb{Z}/2^j\mathbb{Z}$, the value $3k + 2 \pmod{2^j}$ is also uniform over $\mathbb{Z}/2^j\mathbb{Z}$. In particular:

$$P\big(2^j \mid 3k + 2\big) \;=\; \frac{1}{2^j}.$$

Using the standard identity for non-negative integer-valued random variables:

$$E[v_2(3k + 2)] \;=\; \sum_{j=1}^{\infty} P(v_2(3k + 2) \geq j) \;=\; \sum_{j=1}^{\infty} P\big(2^j \mid 3k + 2\big) \;=\; \sum_{j=1}^{\infty} \frac{1}{2^j} \;=\; 1.$$

Therefore $c_5 = E[v_2(3n + 1) \mid n \equiv 5 \pmod 8] = 3 + 1 = 4$.

(4) $r = 7$**:** $3n + 1 = 3(8k + 7) + 1 = 24k + 22 = 2(12k + 11)$. Since $12k + 11$ is odd ($12k$ even, $+11$ odd), $v_2(3n + 1) = 1$ for every k. □

Remark 2.2. Parts (1), (2), (4) are exact pointwise statements: the valuation is a fixed integer on the entire residue class. Part (3) is an expectation statement: the valuation varies over the class $n \equiv 5 \pmod 8$, but its mean is exactly 4. The key tool in part (3) is the invertibility of multiplication by 3 modulo 2^j, i.e. $\gcd(3, 2^j) = 1$, which makes the distribution of $3k + 2 \pmod{2^j}$ exactly uniform.

Remark 2.3. Theorem 2.1 explains the mod-8 step taxonomy of [1]: classes 1, 3, 7 each have a fixed valuation (step types OEE, OE, OE respectively), while class 5 has a variable valuation ≥ 3 (step type OEEE$^+$, at least three even steps).

3 The Balance Identity

Corollary 3.1 (Balance of expected valuations)**.** *The expected valuations satisfy:*

$$c_1 + c_3 + c_5 + c_7 \;=\; 2 + 1 + 4 + 1 \;=\; 8 \;=\; 4 \times 2.$$

The mean expected valuation across all four classes is exactly $2 = c_1$*, the constant valuation of the neutral class* $1 \pmod 8$*.*

Equivalently, in terms of powers of 2 *and* 3*:*

$$\frac{3^4}{2^{c_1} \cdot 2^{c_3} \cdot 2^{c_5} \cdot 2^{c_7}} \;=\; \frac{3^4}{2^8} \;=\; \frac{81}{256} \;=\; \left(\frac{3}{4}\right)^4.$$

Proof. Direct: $2 + 1 + 4 + 1 = 8$ and $2^{c_1} \cdot 2^{c_3} \cdot 2^{c_5} \cdot 2^{c_7} = 2^{2+1+4+1} = 2^8 = 256$. □

Remark 3.2. The balance identity has a clean interpretation in terms of the asymptotic contraction ratios. For large n in class r, the Syracuse step multiplies n by approximately $3/2^{c_r}$:

$$\frac{S(n)}{n} = \frac{3n+1}{n \cdot 2^{v_2(3n+1)}} \longrightarrow \frac{3}{2^{c_r}} \quad \text{as } n \to \infty.$$

The four asymptotic ratios are 3/4, 3/2, 3/16, 3/2 for classes $1, 3, 5, 7$ respectively. Their product is $3^4/2^8 = (3/4)^4$, whose fourth root is 3/4. That is: the *geometric mean* of the four asymptotic contraction ratios equals the ratio of the neutral class 1.

Note that the ratios $3/2^{c_r}$ are exact as limits ($n \to \infty$) for classes 1, 3, 7 where v_2 is constant, and exact as expectations for class 5. The finite-n correction is $O(1/n)$ in each case and does not affect the integer-exponent result.

Remark 3.3 (Asymmetry of the four classes). The four classes play structurally distinct roles:

- Classes 3 and 7 are *growth* classes: $c_3 = c_7 = 1$, asymptotic ratio $3/2 > 1$.
- Class 1 is the *neutral* class: $c_1 = 2$, asymptotic ratio $3/4 < 1$.
- Class 5 is the *contraction* class: $c_5 = 4$, asymptotic ratio $3/16 \ll 1$. A single step in class 5 contributes $c_5 = 4$ to the sum of valuations, the same as *four* steps in classes 3 or 7, or *two* steps in class 1.

The balance $c_1 + c_3 + c_5 + c_7 = 4c_1$ is equivalent to saying class 5's excess valuation $(c_5 - c_1) = 2$ exactly compensates the deficit of classes 3 and 7 combined: $(c_1 - c_3) + (c_1 - c_7) = 2$.

4 Dynamical Implications

4.1 Class 5 as a Dynamical Sink

Theorem 2.1 gives a precise, quantitative sense in which $5 \pmod 8$ acts as a *dynamical sink* in the Collatz tree. Each visit to a $5 \pmod 8$ node contributes an expected valuation of 4 to the denominator of the Syracuse ratio, compared to 2 for class 1, and only 1 for classes 3 and 7. In the asymptotic ratio sense, class 5 contracts by $3/2^4 = 3/16$ per step — four times the exponent of class 1 ($3/2^2 = 3/4$) and sixteen times the inverse of classes 3 and 7 ($3/2^1 = 3/2$).

4.2 Connection to 5-Dominance in Collatz Paths

It is empirically observed that of the first million Collatz sequences, 93.8% pass through the integer 5 on the way to 1, with only 6.2% passing through 32 instead [4]. Theorem 2.1 gives the structural foundation: any orbit entering the $5 \pmod 8$ class experiences a valuation of 4 in expectation, driving it toward 1 far more rapidly than any other class. The complementary probability $6.2\% \approx 1/16$ corresponds to the mod-32 edge density of $5 \to 5$ transitions in the Syracuse graph (see Paper 65 [2]).

4.3 Connection to Steiner Sentence Length Distribution

Paper 65 [2] empirically observes that Steiner sentence lengths follow $P(k) = \frac{1}{3}(3/4)^k$ rather than the naive $(1/2)^k$. The balance identity $c_1 + c_3 + c_5 + c_7 = 4c_1$ is the exact integer expression of the 3/4 decay rate: the neutral class 1 (which governs within-sentence dynamics after conditioning on non-termination) has valuation exactly $c_1 = 2$, giving asymptotic ratio 3/4. The first-principles proof that the within-sentence dynamics produce this exact rate is given in Paper 67 [3].

5 Further Results

Proposition 5.1 (Full distribution for class 5)**.** *For* $n \equiv 5 \pmod 8$ *drawn with natural density,*

$$P\big(v_2(3n+1) = j \;\big|\; n \equiv 5 \pmod 8\big) \;=\; \frac{1}{2^{j-2}}, \qquad j \geq 3.$$

Proof. Writing $n = 8k+5$ gives $v_2(3n+1) = 3 + v_2(3k+2)$ (as in Theorem 2.1(3)). The pmf of $v_2(3k+2)$ under natural density is $P(v_2(3k+2) = m) = 1/2^{m+1}$ for $m \geq 0$: this follows from $P(v_2(3k+2) \geq m) = P(2^m \mid 3k+2) = 1/2^m$ (by the uniformity argument of Theorem 2.1(3)) via $P(v_2 = m) = P(v_2 \geq m) - P(v_2 \geq m+1) = 1/2^m - 1/2^{m+1} = 1/2^{m+1}$. Setting $j = 3+m$ gives $P(v_2(3n+1) = j) = 1/2^{(j-3)+1} = 1/2^{j-2}$. Normalisation: $\sum_{j\geq 3} 1/2^{j-2} = \sum_{i \geq 1} 1/2^i = 1$. $\square$ $\square$

Remark 5.2. This proposition was stated as Open Problem 2 in a draft of this paper. It was machine-verified in Lean 4/Mathlib by the Aristotle automated proof assistant (July 2026); see `ARISTOTLE-REVIEW-66.md` and `Paper66.lean` in the paper directory.

1. **Higher moduli.** The argument for class 5 uses $\gcd(3, 2^j) = 1$ to establish uniform distribution of $3k+2 \pmod{2^j}$. Do analogous exact valuation results hold for finer residue classes modulo 16, 32, and beyond? The mod-32 structure of $5 \to 5$ edges in the Syracuse graph suggests a rich mod-32 refinement.

2. **First-principles sentence length distribution.** The bridge from Theorem 2.1 to a proof of $P(\text{sentence length} = k) = \frac{1}{3}(3/4)^k$ is given in Paper 67 [3].

Acknowledgements

This paper is part of the speculative results stream (64+k) of the Collatz programme. Paper 33 [1] provides the Steiner circuit framework. Paper 65 [2] provides the empirical motivation.

References

[1] J. Seymour, *A Regular Expression Language for the Collatz Graph* (Working Paper, Paper 33), 2026.

[2] J. Seymour, *The* $3^{k-1}/4^k$ *Distribution of Steiner Sentence Lengths* (Working Paper, Paper 65), 2026.

[3] J. Seymour, *First-Principles Derivation of the Steiner Sentence Length Distribution* (Working Paper, Paper 67), 2026.

[4] Reddit post (r/Collatz), *Why does 5 dominate Collatz paths to 1?*, 2026.